Numericals

Date:

Chapter # 06

tf = count of term t in document d

weighted : $TF(t,d) = 1 + \log(tf)$

df = # of docs in the collection that contain the term

$IDF(t) = \log\left(\frac{N}{df_t}\right)$ N = total number of docs

example : N = 1000

df_t = 10 docs (term appears in 10 docs).

$IDF = \log \frac{1000}{10} = 2$

Q. Ranking of Documents

d1 : w1 w2 w4 w6

d2 : w1 w2 w7 w3

d3 : w8 w5 w4 w5

q : w2 w5 w6

steps:

1. vocabulary and df (term | df)
2. TF table (term | docs | q)
3. calculate IDF (term | idf)
4. calculate TF_IDF (same table construct as ②)

optional 5. vector lengths for docs / q.

6. cosine similarity
7. find ranking

step1 : vocabulary & df

term	df
w1	2
w2	2
w3	1
w4	2
w5	1
w6	2
w7	1
w8	1

step2 : calculate TF

	d1	d2	d3	q
w1	1	1	0	0
w2	1	1	0	1
w3	0	1	0	0
w4	1	0	1	0
w5	0	0	2	1
w6	1	0	1	1
w7	0	1	0	0
w8	0	0	1	0

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step 3: calculate IDF

	idf
w1	0.176
w2	0.176
w3	0.477
w4	0.176
w5	0.477
w6	0.176
w7	0.477
w8	0.477

step 4. calculate TF-IDF

	d1	d2	d3	q
w1	0.18	0.18	0	0
w2	0.18	0.18	0	0.18
w3	0	0.48	0	0
w4	0.18	0	0.18	0
w5	0	0	0.96	0.48
w6	0.18	0	0.18	0.18
w7	0	0.48	0	0
w8	0	0	0.48	0

step 6: cosine similarity.

sim(d1, q) = 0.33
sim(d2, q) = 0.08
sim(d3, q) = 0.83

$$sim(a,b) = \frac{a \cdot b}{\sqrt{\Sigma a^2}\sqrt{\Sigma b^2}} = \frac{a \cdot b}{|a||b|}$$

step 7: ranking

d3 > d1 > d2

Jaccard coefficient similarity.

$$similarity = \frac{X \cap Y}{X \cup Y}$$

steps:
1- vocabulary & words
2- calculate similarity using formula.
3- comment on the results.

Q. vocabulary = {multi, label, classification, text, python}
d1 = {multi, label, classification, text}
d2 = {text, classification, python}
d3 = {label, text, classification}
d4 = {classification, label, text}
q = {label, text, classification}

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similarity (d1, q) = $\frac{3}{4}$ = 0.75

similarity (d2, q) = $\frac{2}{4}$ = 0.5

similarity (d3, q) = $\frac{3}{3}$ = 1

similarity (d4, q) = $\frac{3}{3}$ = 1.

The relevance of the retrieved docs against the given query:
d3, d4 (top docs against query)
d1, d2.

Semantic similarity (meaning based similarity)

d1 : mary had little lamb
d2 : little lamb mary had

semantically identical ≈ 1
often written as: 0.99 or 0.999

why not exactly 1?
- vector embeddings rarely match exactly.
- slight structural variation reduces score marginally

so, 0.99 ≈ 1.

important:
- used same words but reordered → ≈ 1
- used synonyms → high (~0.7 – 0.9)
- shares topic only → moderate (~0.4 – 0.6)
- unrelated → near 0.

Q. projection of doc_i on doc_j

$$\text{projection}(doc_i, doc_j) = \frac{doc_i \cdot doc_j}{doc_j \cdot doc_j} \times doc_j$$

Q. compute the Euclidean normalized document vectors

	doc1	doc2	doc3
car	44.33	6.6	39.6
auto	6.24	68.64	0
insurance	0	53.46	46.98
best	21	0	22.5

step1: compute euclidean lengths

$$|doc1| = \sqrt{44.33^2 + 6.24^2 + 0^2 + 21^2} = 49.65$$

$$|doc2| = \sqrt{6.6^2 + 68.64^2 + 53.46^2 + 0^2} = 87.25$$

$$|doc3| = \sqrt{39.6^2 + 0^2 + 46.98^2 + 22.5^2} = 65.43$$

step2: Normalize vectors

$$d1 = \left(\frac{44.33}{49.65}, \frac{6.24}{49.65}, \frac{0}{49.65}, \frac{21}{49.65}\right) = (0.897, 0.126, 0, 0.423)$$

$$d2 = \left(\frac{6.6}{87.25}, \frac{68.64}{87.25}, \frac{53.46}{87.25}, \frac{0}{87.25}\right) = (0.076, 0.787, 0.613, 0)$$

$$d3 = \left(\frac{39.6}{65.43}, \frac{0}{65.43}, \frac{46.98}{65.43}, \frac{22.5}{65.43}\right) = (0.605, 0, 0.718, 0.344)$$